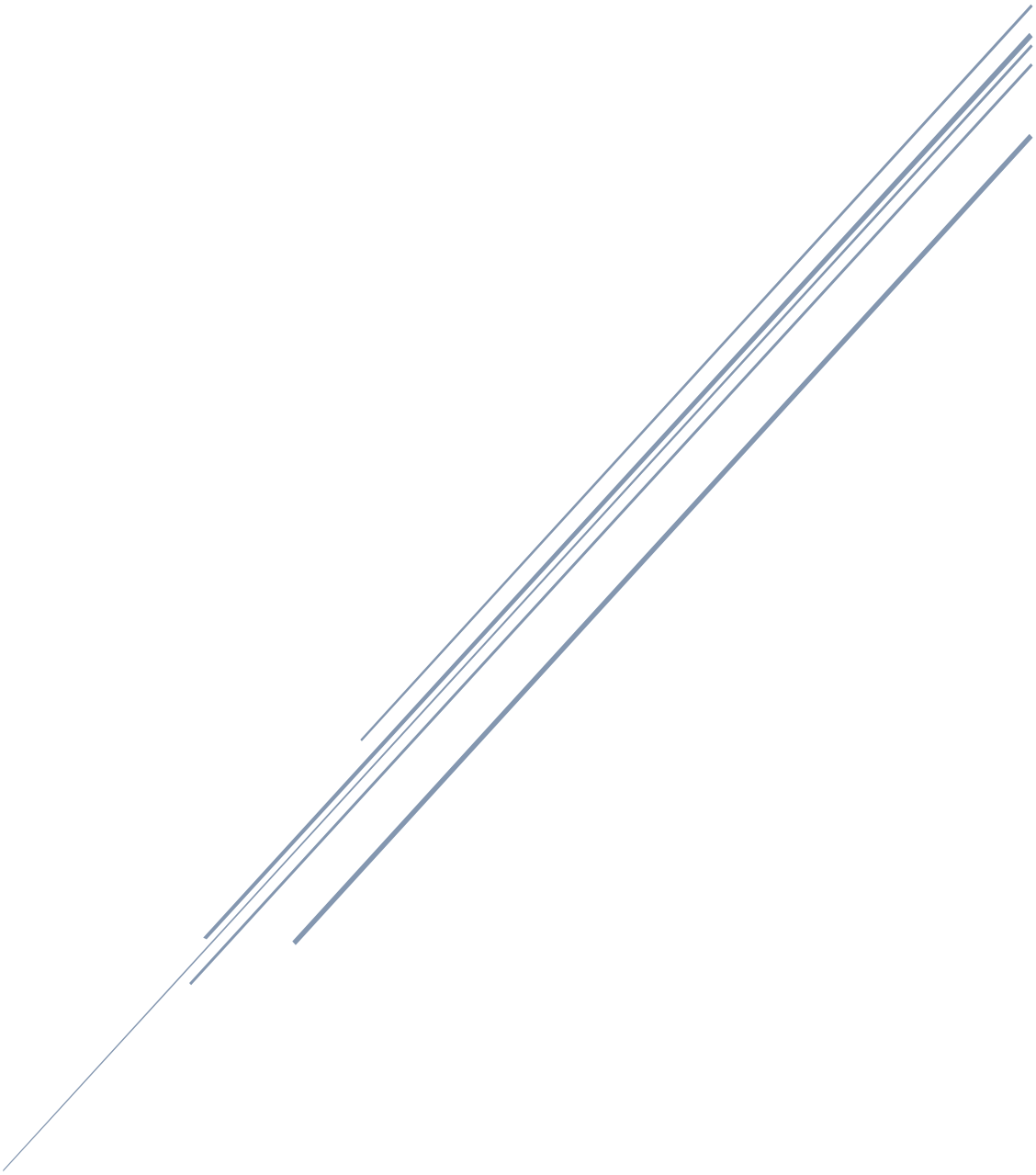
INTERNSHIP DAILY REPORT

Day 1 – Computer Fundamentals & Hardware

CyberShelter Cybersecurity Internship | Week 1



Mohamed Shameem PA
Associate – Information Security (Intern)
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1. Objective

The objective of Day 1 was to understand the fundamentals of computers, including hardware components, memory, storage, CPU architecture, software, operating systems and networking basics, and to explore system hardware, running processes and boot logs using Linux commands in Kali Linux.

2. Lab Environment

- **Host Machine:** Windows PC (AMD Ryzen 5 3450U)
- **Virtualisation:** Oracle VirtualBox
- **Guest OS:** Kali Linux (64-bit), VM name "Kali Lab"
- **Allocated Resources:** 4 GB RAM, 3 CPU cores, 40 GB storage

3. Topics Covered

3.1 Computer – Overview

A computer is an electronic machine that takes data, works on it, and gives us a result. It does five basic things: takes input, stores it, processes it, gives output, and controls the whole process. It is fast and accurate, but cannot think on its own; it only follows the instructions we give it.

3.2 Computer – Components

A computer has three main parts that work together: the CPU, input devices and output devices.

- **CPU:** the brain of the computer that does all the thinking and controls every other part.
- **Memory:** RAM keeps data only while the computer is on; storage (HDD/SSD) keeps it permanently.
- **Input and Output Devices:** input devices give data to the computer; output devices show the result.

3.3 Input Devices

Input devices send data or commands into the computer and convert it into a form the computer can understand.

- **Keyboard:** types letters, numbers and commands using typing, function and special keys.
- **Mouse:** moves the pointer on the screen to click, select and open things.
- **Scanner:** turns a paper document or photo into a digital file.

3.4 Output Devices

Output devices show the result after the computer has processed the data, on screen, on paper or as sound.

- **Monitor:** shows text, images and video using tiny dots called pixels.
- **Printer:** prints output on paper; laser and inkjet are the most common.
- **Speakers:** give output as sound, like music, videos or alerts.

3.5 Data Storage & Memory

Memory (RAM) keeps the data we are using right now, while storage keeps our files saved permanently.

- **RAM:** very fast but temporary; loses everything when the power is off.
- **HDD:** saves data on spinning disks; cheap and large, but slower.
- **SSD:** saves data on memory chips with no moving parts; much faster but costs more.

3.6 Memory Units

Memory units tell us how much data can be stored. Computers understand only 0s and 1s, so every unit starts from the bit.

- **Bit:** the smallest unit of data, either 0 or 1.
- **Byte:** 8 bits together; stores one character, like the letter "A".
- **Bigger Units:** 1 KB = 1,024 Bytes, 1 MB = 1,024 KB, 1 GB = 1,024 MB, 1 TB = 1,024 GB.

3.7 Types of Data Storage

Data storage is divided into types based on how fast it is and how long it keeps data.

- **Primary Storage:** RAM; fast but temporary.
- **Secondary Storage:** HDD, SSD or pen drive; permanent storage for files, apps and the OS.
- **Tertiary Storage:** magnetic tapes and DVDs used for long-term backup and archives.

3.8 Computer Architecture – CPU, ALU and CU

Computer architecture is how the main parts inside the CPU work together to run programs.

- **CPU:** runs every instruction; speed is measured in GHz and modern CPUs have multiple cores.
- **ALU:** does calculations (+ − × ÷), comparisons and logic like AND, OR and NOT.
- **CU:** works like a manager using the fetch–decode–execute cycle.

3.9 Software

Software is a set of programs that tells the computer what to do. Without it, hardware is useless.

- **System Software:** runs the computer and connects hardware with the user (Windows, Linux, Kali).
- **Application Software:** helps the user do a specific job (MS Word, Excel, Chrome).

3.10 Operating System

An operating system manages the whole computer and sits between the user and the hardware. Its main part is the kernel. Examples: Windows, Linux, macOS, Android.

- **Process Management:** decides which program uses the CPU and for how long.
- **Memory Management:** gives RAM to programs and takes it back when they close.
- **Security:** protects the system using passwords, permissions, updates and logs.

3.11 Networking Basics

A network is two or more devices connected to share data and resources, using cables or Wi-Fi.

- **LAN:** a small network in one place, like a home or office.
- **WAN:** connects networks across cities or countries; the internet is the biggest WAN.
- **Network Devices:** router (connects to the internet), switch (connects devices in a LAN), NIC (network card with a unique MAC address).

4. Tasks Performed

All commands were executed in the Kali Linux virtual machine using the terminal.

4.1 Understand the System

4.1.1 CPU Information

- **Command Executed:** `lscpu`
- **Purpose:** Displays detailed information about the processor.
- **Observation:** The CPU is an **AMD Ryzen 5 3450U with Radeon Vega Mobile Gfx**, running on x86_64 (64-bit) architecture. **3 CPU cores** are allocated to the virtual machine, and the output confirms the system runs under a hypervisor with full virtualisation.



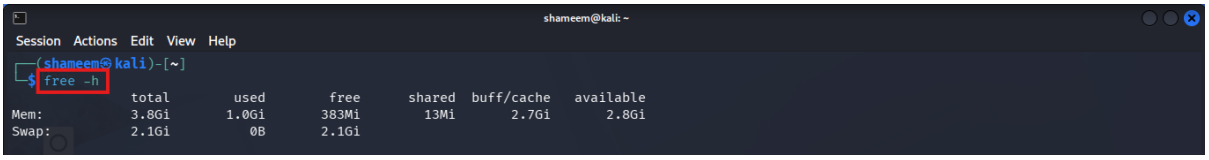
CPU model, cores and architecture.



CPU cache and security details.

4.1.2 Memory Usage

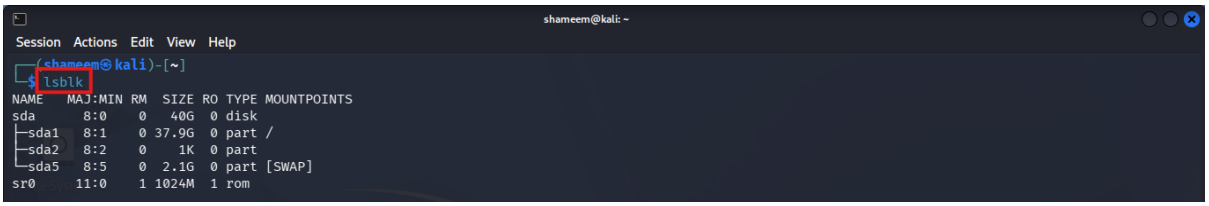
- **Command Executed:** `free -h`
- **Purpose:** Displays total, used and available RAM and swap in a human-readable format.
- **Observation:** Total RAM is **3.8 GiB**, with about 1.0 GiB used and 2.8 GiB available. Swap memory is 2.1 GiB and is currently not in use.



RAM and swap usage.

4.1.3 Disk Partitions

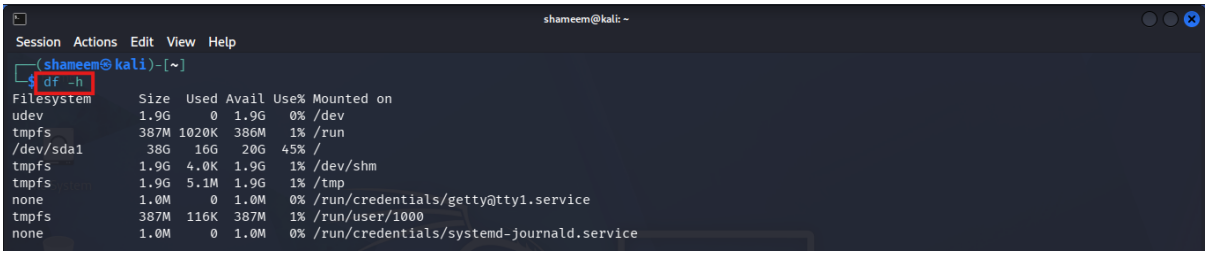
- **Command Executed:** `lsblk`
- **Purpose:** Lists all storage devices and their partitions.
- **Observation:** One **40 GB disk (sda)** divided into sda1 (37.9 GB, root /), sda2 (extended) and sda5 (2.1 GB, swap). sr0 is the virtual CD/DVD drive.



Disk and partition layout.

4.1.4 Disk Space Usage

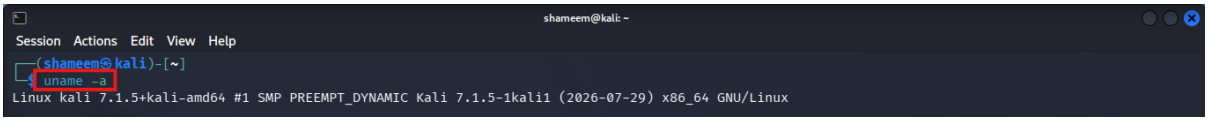
- **Command Executed:** `df -h`
- **Purpose:** Displays used and available space on each mounted file system.
- **Observation:** The main partition **/dev/sda1** is 38 GB, with 16 GB used and 20 GB available (45% used).



File system usage.

4.1.5 System and Kernel Information

- **Command Executed:** `uname -a`
- **Purpose:** Displays the operating system, kernel version, hostname and architecture.
- **Observation:** The system runs a **Kali Linux kernel on x86_64** architecture, with the hostname kali.

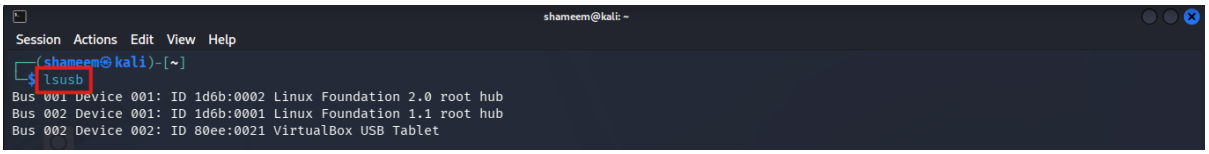


Kernel and system information.

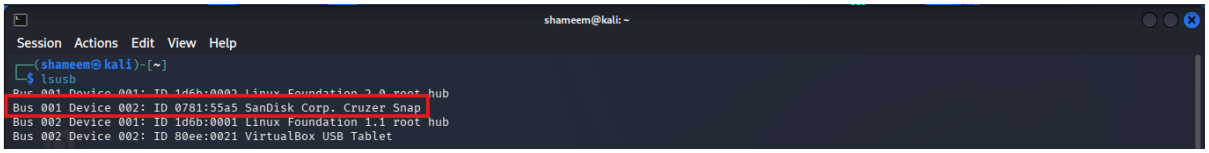
4.2 Explore Hardware

4.2.1 USB Devices

- **Command Executed:** `lsusb`
- **Purpose:** Lists all USB devices connected to the system.
- **Observation:** Without an external device, the output shows two USB root hubs (USB 1.1 and 2.0) and the VirtualBox USB Tablet (mouse integration). After connecting a pen drive, a new device was detected: **SanDisk Corp. Cruzer Snap**, showing how Linux detects USB devices as soon as they are connected.



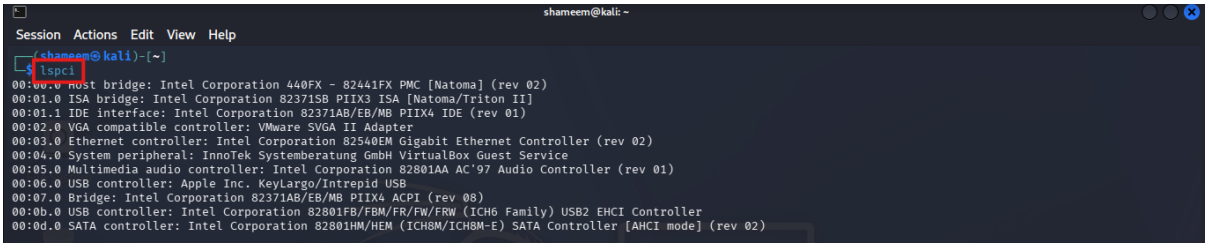
USB devices before connecting the external drive.



SanDisk Cruzer Snap detected after connecting the drive.

4.2.2 PCI Hardware Devices

- **Command Executed:** `lspci`
- **Purpose:** Lists internal hardware connected to the motherboard through the PCI bus.
- **Observation:** Shows the virtual hardware provided by VirtualBox, including the VGA graphics adapter, **Intel 82540EM Gigabit Ethernet** controller (network card), AC'97 audio, USB controllers and the SATA controller.

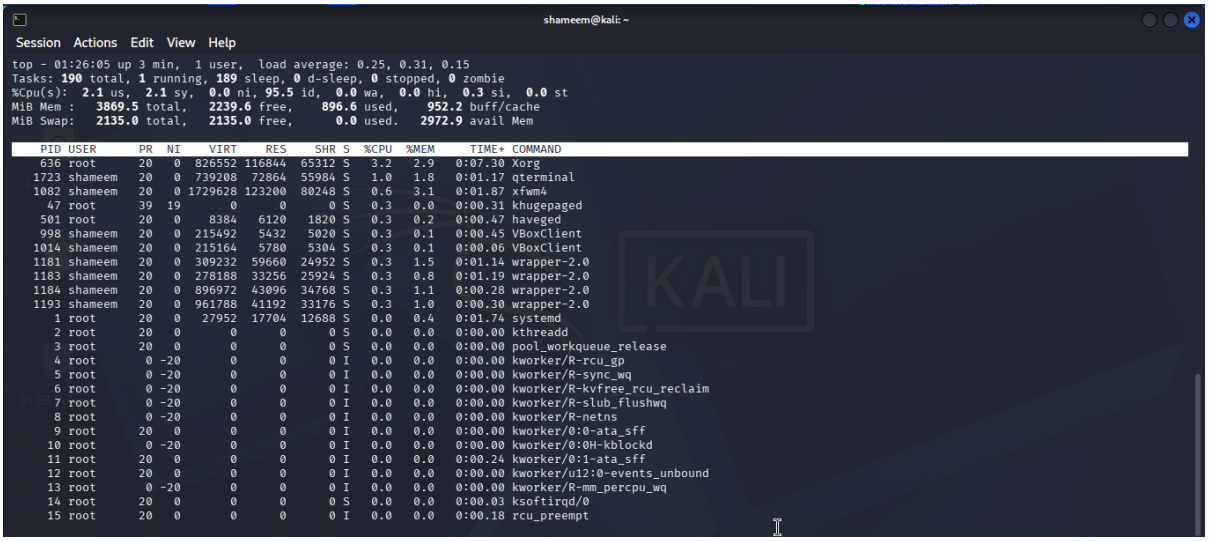


Internal PCI hardware.

4.3 Process Basics

4.3.1 Running Processes

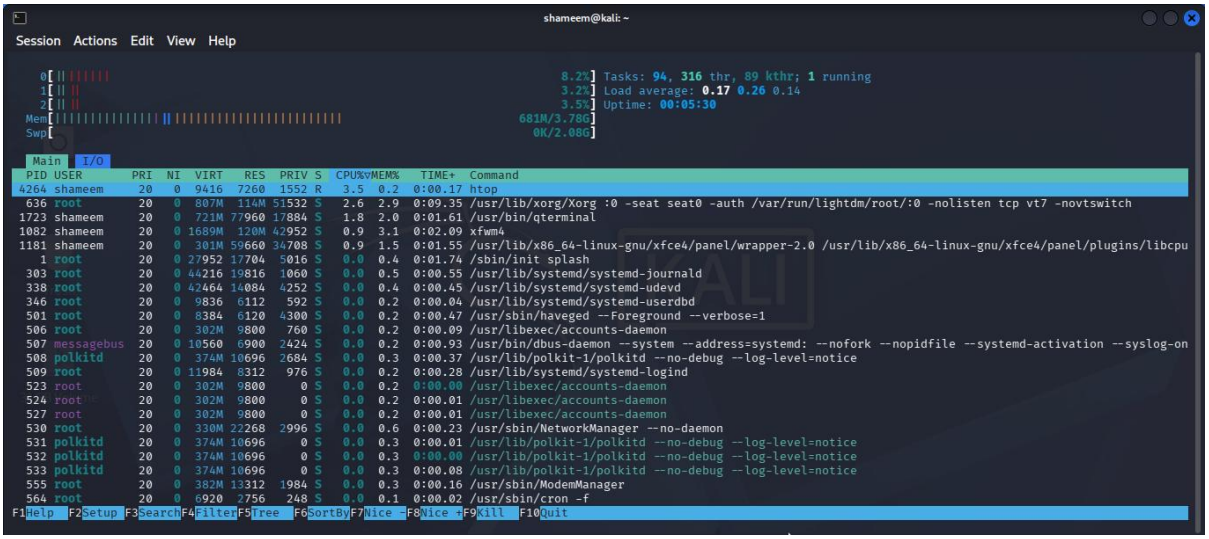
- **Command Executed:** `top`
- **Purpose:** Displays a live view of running processes with CPU and memory usage.
- **Observation:** Refreshes every few seconds and shows the processes using the most CPU and memory, along with uptime, number of tasks and load average.



Live process monitoring with top.

4.3.2 Interactive Process Viewer

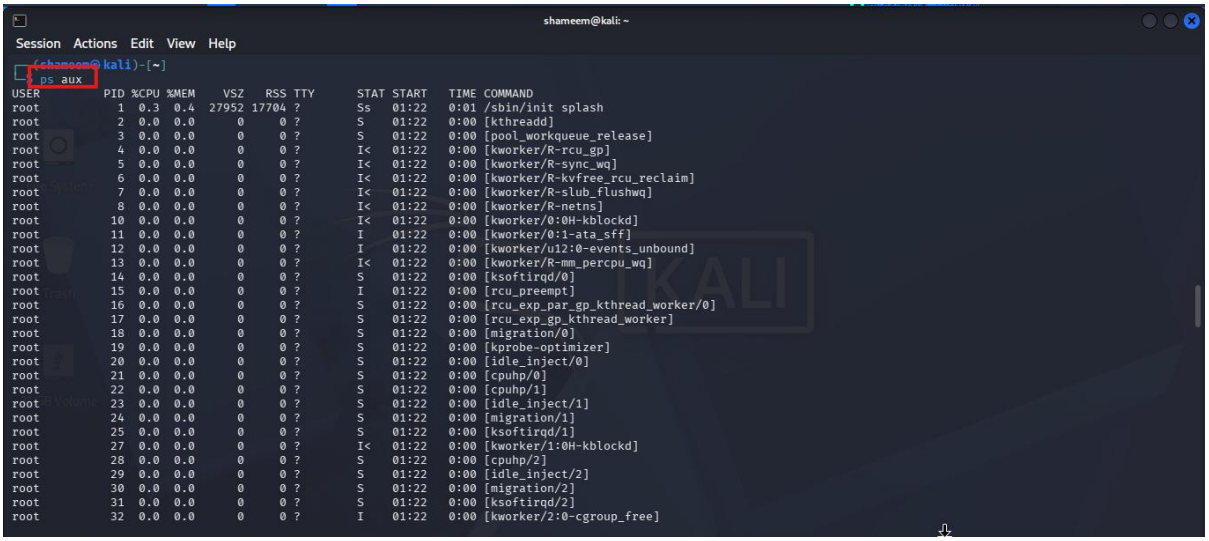
- **Command Executed:** `htop`
- **Purpose:** An interactive, easier-to-read version of top.
- **Observation:** Displays CPU cores, memory and swap usage as colour bars, and allows scrolling, searching and managing processes.



Process monitoring with htop.

4.3.3 Process List

- **Command Executed:** `ps aux`
- **Purpose:** Displays a snapshot of all processes running at that moment.
- **Observation:** Each process is listed with its user, PID, CPU %, memory %, start time and command. **PID 1 (/sbin/init)** is the first process started by the kernel at boot.

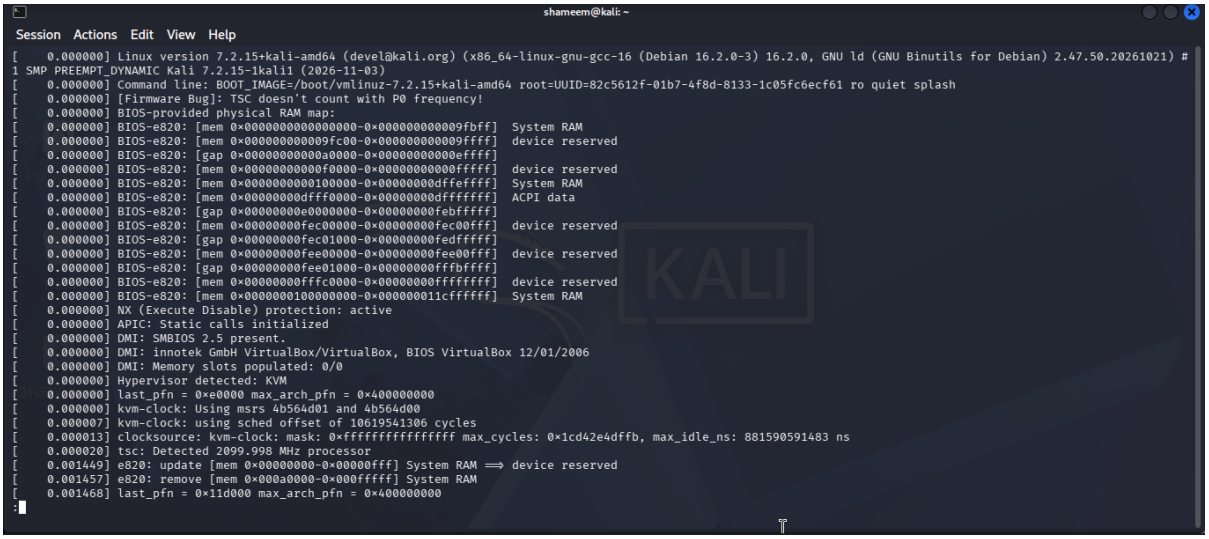


Snapshot of all running processes.

4.4 Boot & Logs

4.4.1 Kernel Boot Messages

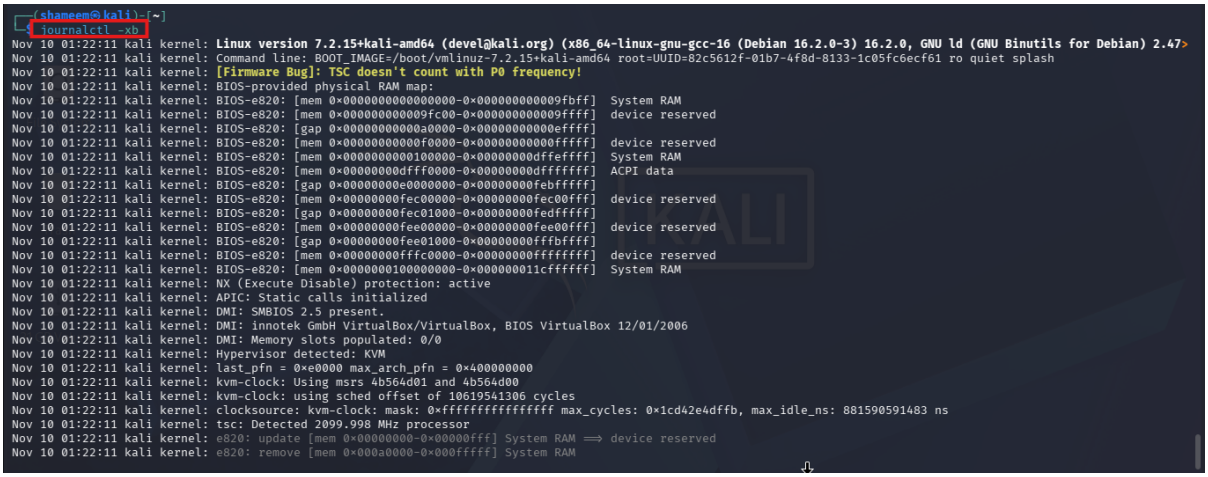
- **Command Executed:** `sudo dmesg | less`
- **Purpose:** Displays kernel messages, mainly about hardware detection during boot.
- **Observation:** Shows the kernel detecting and initialising the CPU, memory, disks, USB and network devices step by step during startup.



Kernel ring buffer messages.

4.4.2 System Boot Logs

- **Command Executed:** `journalctl -xb`
- **Purpose:** Displays system logs for the current boot, with additional explanations.
- **Observation:** Shows the kernel version, boot parameters, memory map and hardware detection, followed by services started by systemd. It also confirms the VM environment (**Hypervisor detected: KVM**).



System journal for the current boot.

5. Results

5.1 Hands-on Task Findings

Item	Result
CPU Model	AMD Ryzen 5 3450U with Radeon Vega Mobile Gfx (3 cores allocated)
RAM Size	3.8 GiB (approx. 4 GB), with 2.1 GiB swap
Disk Partitions	sda 40 GB – sda1 37.9 GB (root /), sda2 (extended), sda5 2.1 GB (swap)
USB Devices	USB 1.1 and 2.0 root hubs, VirtualBox USB Tablet, SanDisk Cruzer Snap (when connected)

5.2 How a System Boots

1. **Power On:** The power supply provides power to all components of the computer.
2. **BIOS/UEFI:** The firmware runs POST (Power-On Self-Test) to check the CPU, RAM and disks.
3. **Bootloader (GRUB):** The firmware finds GRUB on the disk, which loads the Linux kernel into memory.
4. **Kernel:** The kernel detects hardware and loads drivers; these messages are visible in `dmesg`.
5. **systemd (init):** The kernel starts systemd (PID 1), which starts all system services; visible in `journalctl -xb`.
6. **Login:** The login screen appears and the user can start using the system.

6. Key Learnings

- Understood how the CPU, memory, storage and input/output devices work together to process data.
- Learned to identify system hardware in Linux using `lscpu`, `free -h`, `lsblk`, `lsusb` and `lspci`.
- Learned how to monitor running processes with `top`, `htop` and `ps aux`.
- Understood the boot process and how to read boot logs with `dmesg` and `journalctl`, which are important for troubleshooting and security investigation.
- Observed in real time how Linux detects a new USB device, which is useful for monitoring unauthorised devices.

7. Challenges Faced

- **Reading long outputs:** Commands like `dmesg` and `journalctl` produce very long logs. Using `less` and scrolling helped to read them step by step.
- **Kernel version change:** After running the system update, the kernel version changed, so the outputs of `uname -a` and `journalctl` needed to be captured again after a reboot to stay consistent.

8. Conclusion & Next Steps

Day 1 built a strong foundation in how a computer works, from its main components to the operating system and networking. The practical session showed how Linux commands can be used to inspect hardware, monitor processes and read boot logs, skills that are essential for system analysis and security work.